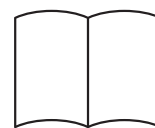




Operator's and Maintenance Manual

Sentry® Horizontal Shaft Impactor

WO# 2254830
MODEL: F4355IHSC
11-2022



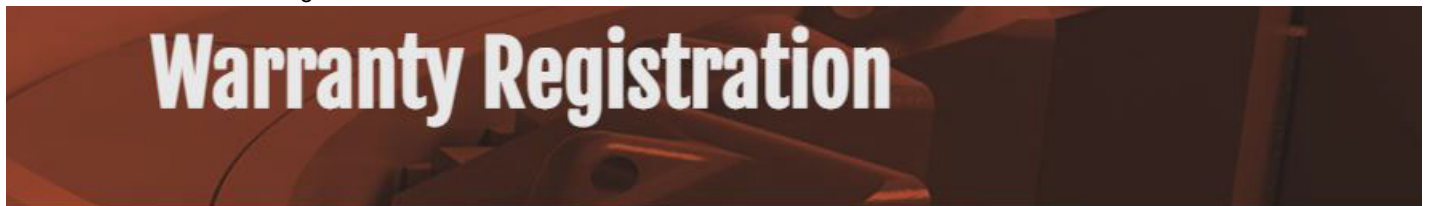
WARNING!
Read manual before
operating equipment.

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Warranty Registration

1. Go to <https://superior-ind.com/customer-support/warranty/>
2. Complete warranty form to register your equipment with our customer service department. Completion ensures that equipment will be covered under the seller's standard warranty. Read through standard warranty located in the back of this manual.
3. Unit Serial Number and model can be found on front cover of this manual. Unit Serial Number can also be found on conveyor nameplate.
4. After all information is entered, check box agreeing with bolded statement.
5. Click submit to finish registration.



Please complete the form below to register your equipment with our customer service department. The completion ensures the equipment will be covered under the seller's standard warranty, which is located in the back of your owner's manual.

On behalf of Superior's hard-working employees and our great network of dealers, thank you for your purchase!

Name *

First

Last

Serial Number

Model

Service Start Date

Transaction Type

- ☐ Sale
☐ Lease
☐ Rental
☐ Change of Ownership

Are you a dealer or owner/lessee? *

- ☐ Dealer
☐ Owner/Lessee

Next

Foreword

This instructional manual was compiled for the benefit of all those involved in the maintenance and operation of Superior Industries products. When carefully read, and suggested operating and service procedures applied, the information contained within this instruction manual can save valuable time and eliminate broad assumptions.

A careful study by everyone working on the equipment should be made of this manual to familiarize yourself with the mechanical make-up and design features of the product before it is placed in operation.

It contains important instructions for safe, optimal and economical use of the equipment. Follow these instructions to avert possible hazards, cut repair costs and the number of failures, increase the equipment's reliability and prolong its service life

This manual should be completed with local safety and environmental protection regulations. In addition, the generally accepted technical rules governing safety in the workplace should be followed.

A copy of this user manual should always be available near the equipment.

The descriptions and specifications in this manual were accurate when it was approved for publishing.

Superior Industries reserves the right to discontinue it's models at any time and change the specification or drawings, without notice or any obligation.

Should you have any questions regarding the equipment or this publication, please contact Superior Industries.

Contact Us

<https://superior-ind.com/contactus/>

Product Support

- Available 24 hrs / 7 days a week / 365 days a year.
- +1 (320) 589-7485
- productsupport@superior-ind.com

Parts Sales

- Available 24 hrs / 7 days a week / 365 days a year.
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HSI Safety & Guidelines



Read and understand equipment operator's manual before operating or performing maintenance. Failure to do so could result in serious injury or death.

Signal Words and Symbols



This is the safety alert symbol. It is used to alert you to potential physical injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.



DANGER indicates a hazardous situation which, if not avoided, will result in death or serious injury.



WARNING indicates a hazardous situation which, if not avoided, could result in death or serious injury.



CAUTION indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.



NOTICE is used to address practices not related to physical injury.



Safety instructions (or equivalent) signs indicate specific safety related instructions or procedures.

Superior Industries cannot anticipate every possible circumstance which involves potential hazard. The warnings and notifications in the manual are not all inclusive.

Owner Responsibilities

- Operate and maintain equipment in a safe manner and in accordance with all applicable local, state, and federal codes, regulations and/or laws; and in compliance with equipment safety labeling and manual instructions.
- All personnel must read this manual and thoroughly understand safe and correct installation, operation, and maintenance procedures.
- Install equipment correctly before placing in service and then follow recommended maintenance schedule and procedures to service equipment thereafter.
- Fulfill all warranty obligations so as to not void warranty. Complete description of obligations is found on back of sales contract under terms and conditions. Warranty policy is located on inside back cover of this manual.

Safety Rules

These are general safety considerations. Additional precautions may be necessary to operate your equipment in a safe manner.

- Follow all applicable local, state, and federal codes, regulations and/or laws; OSHA rules and regulations; insurance requirements; and in compliance with equipment safety labeling and manual instructions in regards to operating and maintaining your equipment.
- Do not allow anyone to operate equipment until he/she has read manual and is familiar with safety precautions.
- Do not allow unauthorized persons unfamiliar with equipment operate or perform maintenance.
- Do not operate or maintain equipment while under the influence of alcohol, medications, or other drugs which can impair judgment or cause drowsiness.
- Required PPE (personal protective equipment) will vary depending on equipment usage and location. It is operator's responsibility to wear correct PPE for the situation and when required.
- It is operator's responsibility to be aware of hazardous areas and follow safety signs and label instructions.
- It is operator's responsibility to be aware of equipment operation and work area hazards at all times. Keep work areas clear and clean.
- Loose clothing, long hair or jewelry can get caught in moving parts. Remove or restrain when necessary.
- Do not smoke while servicing the equipment.
- Do not perform maintenance while equipment is running unless noted otherwise in a procedure within this manual.
- Lockout/Tagout/Blockout equipment before performing maintenance.
- It is operator's responsibility to know all locations, and periodically check, all guards and shields are in place before operating equipment.
- Watch for adequate overhead clearance and power lines before setting up and operating equipment.
- Operators are responsible to know location and function of all controls and indicators.
- Be sure area is clear of personnel before starting equipment for operation or maintenance.
- Equipment requires an operator at all times. Never leave machine running unattended.
- The equipment should not be used to handle materials other than what it was designed for. Refer to specifications.

- Do not use equipment if safety devices are not working properly. Remove from service until repaired.
- Modifying equipment or using unapproved factory recommended service parts may result in injury, death, voided warranty, and/or decrease equipment effectiveness.
- All personnel must be trained in safe use of open flame heating devices.
- Working with cryogenics such as liquid nitrogen and dry ice, presents unique safety hazards. Personnel must be trained in safe use of these agents and appropriate PPE.
- All personnel must be trained in proper use of compressed air tools.
- Personnel must be trained in the use of torque multipliers.
- Only trained and certified welders are to perform welding tasks.

Fall Hazards

Working with / around crusher can present a number of potential slip and trip hazards. Rock spillage and grease leaks can cause slip and trip hazards. Interior of eccentric assembly, and gear well areas will be wet with lubricating grease, presenting a serious slip hazard.

Follow guidelines below:

- Clean up all grease spills that may drain off on major crushing components that have been removed from crusher, or that may have developed from system leaks.
- Use extra care when entering grease soaked areas of crusher. Use care in keeping this area clean. Ensure grease is collected and disposed of properly.
- Ensure rock spillage is cleaned up promptly.
- Use caution when using walkways. Wet crusher feed conditions and excessive dust suppression water sprays can cause wet fines to accumulate on walkways.

Flying / Falling Object

Overflow, worn chutes, or other possibilities for spillage can cause falling rock to present a safety hazard. Crushing action can cause rock to be ejected from crushing cavity, particularly as rock is first being introduced into cavity.

Follow guidelines below:

- Safety skirt, normally made of flexible rubber, can be mounted between feed chute and crusher feed hopper to keep rock from being ejected or from spilling out if crusher cavity becomes overfilled. Skirt also serves as part of the system to control fugitive dust from crusher.

- Spillage can accumulate on lower deck and present slip and trip hazards. Good housekeeping is required for safe crusher operation.
- Work should be coordinated to eliminate potential for dropped tools and parts, areas under workers cordoned off, or adequate overhead protection should be provided.

Confined Spaces

Crusher discharge area is normally considered a confined space and should be treated as such. Routine maintenance procedures requires visual inspection of underside of crusher, crusher lubrication lines, and discharge cavity.

Following concerns must be addressed:

- Confined space work permits and procedures must be utilized whenever maintenance work is to be carried out under crusher or in discharge cavity.
- Electrically isolate and lockout feed conveyor, crusher, and discharge conveyor.
- Discharge area must be designed in such a way as to provide safe and easy access to area under crusher. Discharge area must be designed to eliminate spillage and allow for adequate dust control.

Respiratory Concerns

Crushing of rock produces rock dust, which is often fine enough to be carried into lungs during normal breathing. All rock dust presents some health hazards, with some rock presenting especially serious health hazards. System must be provided to control escape of rock dust, or PPE must be provided to protect workers.

Following items must be addressed:

- Seriousness of dust control problem can not be determined until after crusher has been in operation for short time.
- Working in discharge area of crusher may agitate / kick up dust.

High Noise Levels

Crushing rock generates high level of noise. Typical decibel readings are in the 85 - 105 dB range. Use of noise protection devices are required.

Below is a list of concerns:

- Actual noise level of crusher needs to be determined by measurement during normal crushing operations. Once noise level is measured, proceed to determine the proper PPE required for that level of noise. Time exposure limits may also need to be put in place.
- Exceeding crusher limits can generate extremely high noise levels. Only operate crusher within design limitations.

General Housekeeping

Follow guidelines below:

- Always correct causes of spillage as they occur and clean up spillage as soon as possible.
- Clean up all grease spills or other types of spills that would result in slip hazards.
- Pick up and put away tools, cable rigging, wash down water hoses, air pressure hoses, and acetylene torch equipment immediately after they are used.

Proper Training and Indoctrination

All personnel must be properly trained and receive proper hazard awareness training.

The following are topics of job specific training:

- All personnel must become familiar with general operation and maintenance of crusher.
- All personnel must be made aware of specific safety hazards related to working around crusher.
- All personnel must be trained in safe use of tools and materials used in installation, maintenance, and operation of crusher.

Support Structure

- Vibratory equipment support structure must comply with and achieve performance set by ISO 2372 standard.
- Any vibratory equipment (e.g., vibratory screens) supported on a support structure that is not in compliance with ISO 2372 standard must not be operated.
- Operation of any Superior Industries vibratory equipment on a structure that does not meet ISO 2372 standard may result in damage to equipment, damage to support structure, and serious bodily injury.
- Applicable vibration severity range maximum limit for this equipment to comply with ISO 2372 is 0.62 in/s (11.2 mm/s).
- If you do not have access to ISO 2372 or are unsure how to comply with ISO 2372 or these requirements, you must contact Superior Industries before operation of vibratory equipment.
- Failure to use support structure in compliance with the ISO 2372 standard (or to comply with the provisions of preceding paragraph) will void any applicable warranty on any Superior Industries equipment used with such non-compliant support structure.
- Vibration measurements should be taken directly under each spring base/s and recorded in your warranty documentation and maintenance file.

Lockout/Tagout/Blockout



Read and understand equipment operator's manual before operating or performing maintenance. Failure to do so could result in serious injury or death.

⚠ WARNING

Failure to follow instructions below could result in death or serious injury:

- Follow correct Lockout/Tagout/Blockout procedures.
- Follow all applicable local, state, and federal codes, regulations and/or laws; OSHA rules and regulations; insurance requirements; and in compliance with equipment safety labeling and manual instructions in regards to operating and maintaining your equipment.
- Use proper tools to remove unconstrained material; do not remove with hand or other body parts which may become trapped.

Lockout/Tagout

Place a lock and tag(s) on an energy isolating device, in accordance with a procedure, ensuring that the energy isolating device and equipment being controlled cannot be operated until lock and tag(s) are removed.

Lockout/Tagout/Blockout:

- Protects against unexpected or accidental start up or movement of equipment and/or parts.
- Notifies other personnel when equipment is unsafe to operate.
- To prevent injury from stored energy in equipment even when it is shutdown.

Procedure

Figure 1



1. Identify energy source(s).

2. Isolate energy source(s).
3. Lock and Tag energy source(s).
4. Prove equipment isolation is effective.

According to 29CFR part 1910 of OSHA regulations, the employer must establish a Lockout/Tagout system of procedures, training, and periodic inspection before any employee operates, services, or maintains a piece of equipment.

Follow employers policies and recommendations for Lockout/Tagout procedures.

Lockout/Tagout when:

- Repairs or maintenance is required on equipment.
- Cleaning or lubricating equipment.
- Cleaning blocked or jammed mechanisms.
- Equipment is left unattended.

Lockout/Tagout Rules:

- An authorized maintenance person/ foreman/ team leader/ anyone working on equipment can apply a lock and tag to equipment.
- Each person working on equipment applies their own tag.
- Only the person who applied lock and tag can remove it.

Blockout

In addition to Lockout/Tagout, Blockout prevents movement resulting from stored energy when a belt becomes stalled.

A conveyor belt may be under high tension and could move even when system is locked out.

Because fugitive material causes a motor overload this creates a tension on the belt between the drive pulley and the wedge causing stored energy.

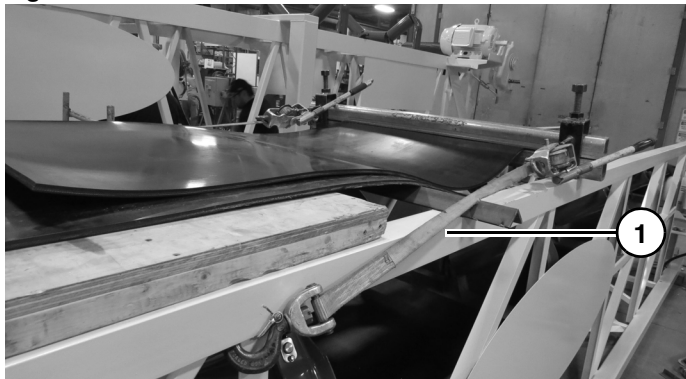
To prevent the release of stored energy “blockout” the belt by clamping in such a way as to prevent movement and possible entrapment.

Procedure

Figure 2



Figure 3



1. Place belt clamp on belt (*Figure 2*).
2. Remove belt tension by pulling chain or cable come along in opposite direction of belt travel (1) (*Figure 3*).

General Information



Read and understand equipment operator's manual before operating or performing maintenance. Failure to do so could result in serious injury or death.

⚠ WARNING

Failure to follow instructions below could result in death or serious injury:

- Lockout/Tagout before performing maintenance and operation.
- Follow installation instructions completely. Failure to do so may cause death or serious injury.

Foundation

NOTICE

- Foundation must be durable, solid, and level in order to safely operate crusher.

Lifting Crusher

⚠ WARNING

Failure to follow instructions below could result in death or serious injury:

- Lifting crusher onto foundation should only be accomplished using lifting hooks.

Clearance Dimensions

- Room is needed under crusher for discharge conveyor and / or chute.

Feed Arrangement

NOTICE

- Inadequate amount of feed material will cause crusher to run underpowered, resulting in uneven wear, potentially causing damage to equipment.
- Uneven feed distribution will result in oversize product and uneven wear.

Use of level sensor is recommended to monitor feed level.

- Feed size should be thoroughly mixed and evenly distributed around crushing cavity.

- Feed box mounted over crusher can serve as backstop to ensure material falls vertically.
- Feed spout mounted on feed box and centered over feed plate will aid in providing uniform distribution of feed around crushing cavity.

Discharge Arrangement

- Replaceable wear material should be used to line discharge compartment.
- Within discharge compartment, a shelf / ledge should be incorporated to make a "dead bed" for crusher discharge material to fall on.
- Dead bed is designed to absorb most of energy of falling material before cascading onto conveying equipment.
- Discharge chute must be positioned at 45° angle or greater from horizontal. If wet discharge material is present, chute angle should be increased.
- Maintain adequate clearance between discharge equipment and crusher.

Lube Unit

NOTICE

- Grease system has gravity drain and should not have any uphill runs in it.
- High points could cause air lock and prevent grease from returning to tank.
- Do not run lubrication lines more than 6 meters (20 feet).

- 2 HP Lube Pump Drive Motor
- Located near lid lifter.

Note: Check grease level in hydraulic tank frequently during first few days of operation to ensure grease is not leaking out of pedestal.

Note: If grease needs to be added continuously, check for leak in system. Contact Superior Industries if problem is not resolved.

Sentry® HSI Decal Layout

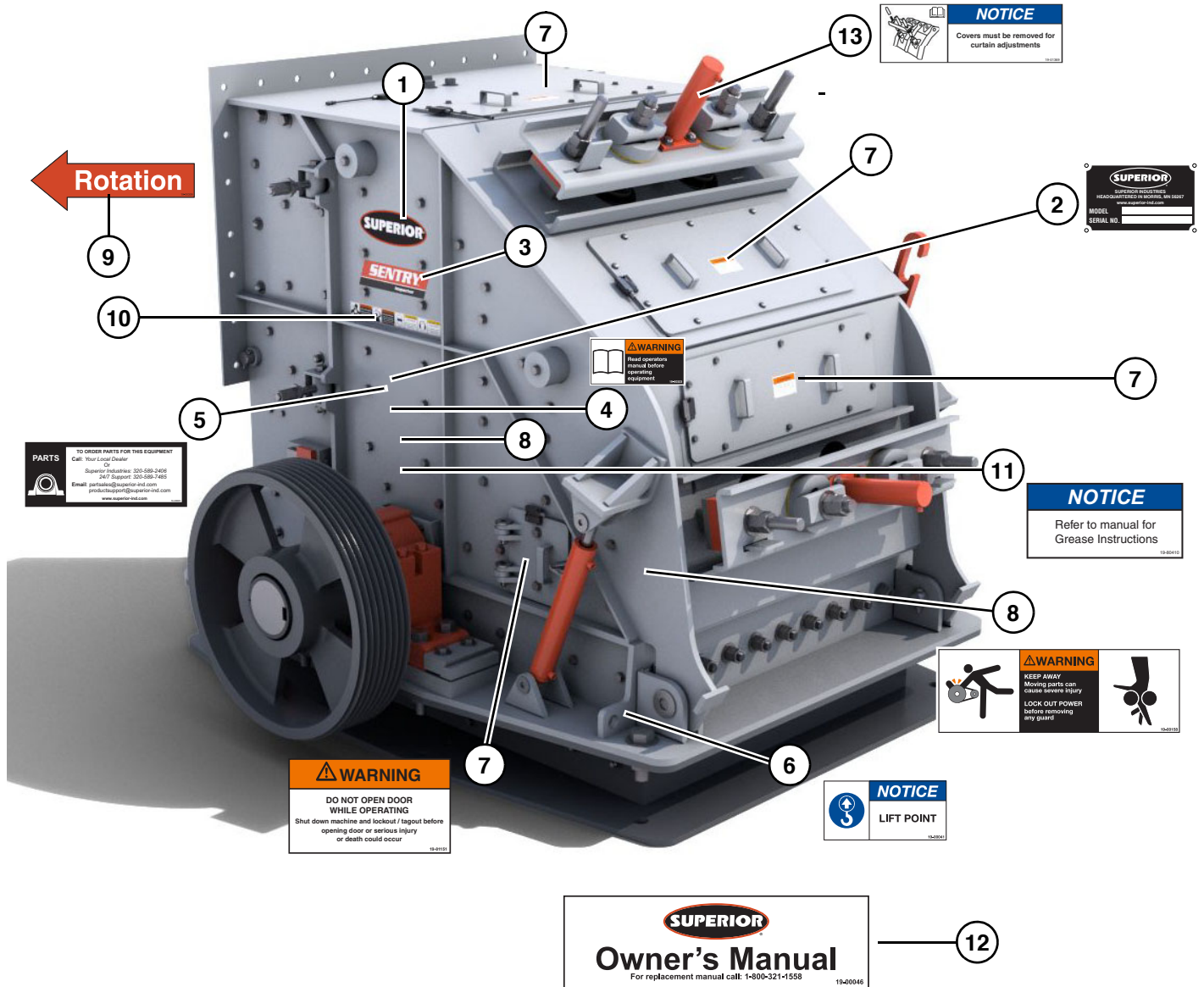


Read and understand equipment operators manual before operating or performing maintenance. Failure to do so could result in serious injury or death.

Note: Inspect decals every time equipment is cleaned, keep free of debris and replace if decal becomes faded, worn or is peeling.

Right Side

Figure 4



Refer to (Figure 4) for the following:

1. 19-00012 Superior Logo 15" x 5"
 - Each side of equipment, Qty-2
2. 19-01104 Serial Number Plate
 - Side of equipment, Qty-1
3. 19-80403 Sentry® HSI Logo
 - Each side of equipment, Qty-2
4. 19-00323 Read Operator's Manual
 - Side of equipment, Qty-1
5. 19-00081 Parts Order
 - Side of equipment, Qty-1
6. 19-00041 Notice Lift Point
 - Next to lift points on corners, Qty-4
7. 19-01151 Do Not Open
 - On access doors, Qty-5
8. 19-00158 Moving Parts
 - Each side of equipment, Qty-4
9. 19-00328 Rotation Arrow
 - Left side of equipment, Qty-1

Note: Left side is not shown in previous picture.

Note: Crusher rotates clockwise in this image

10. 19-01136 Safety Strip
 - Each side of equipment, Qty-2
11. 19-80410 Refer to Manual for Grease Instructions
 - Each side of equipment, above bearings, Qty-2
12. 19-00046 Owner's Manual Decal
 - On owner's manual compartment, Qty-1
13. 19-01369 HSI Curtain Caps
 - On caps, Qty-varies

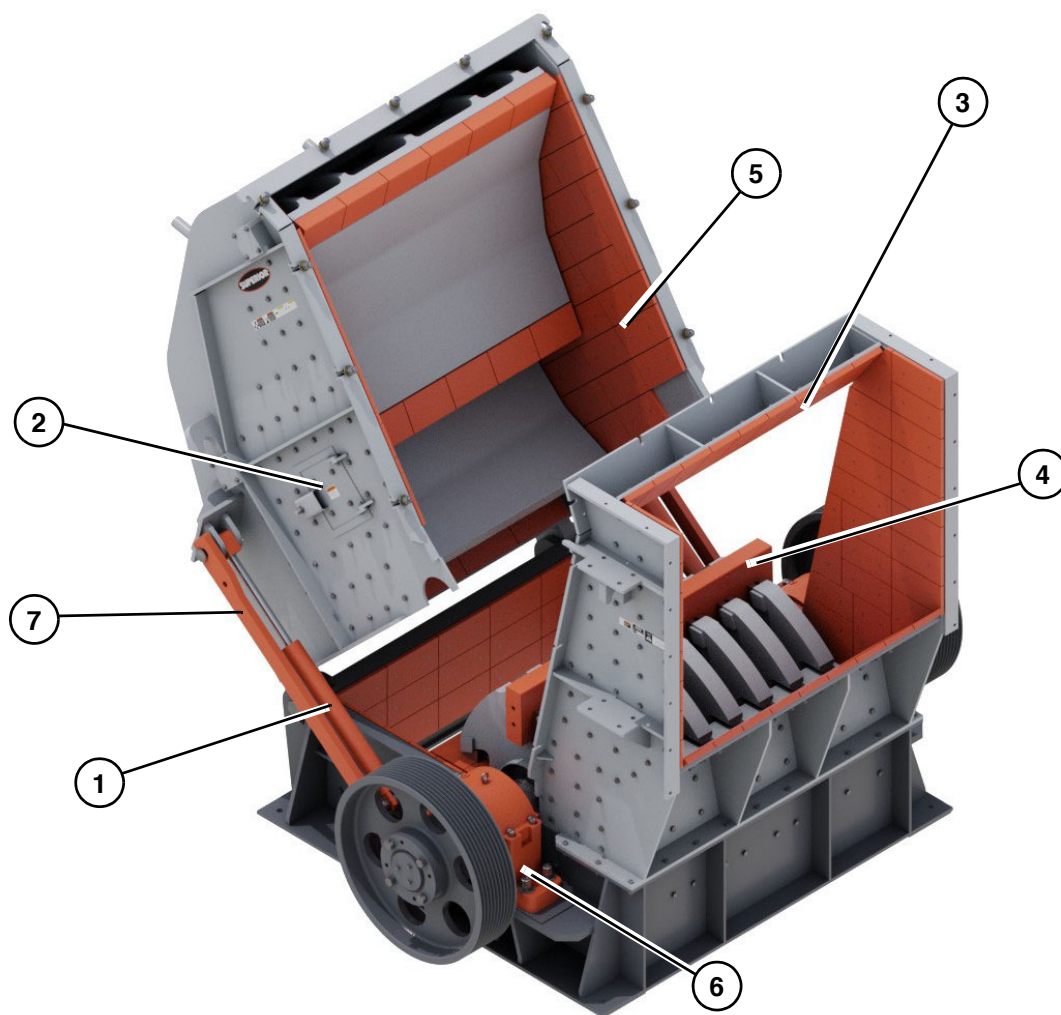
General Assembly Overview



Read and understand equipment operator's manual before operating or performing maintenance. Failure to do so could result in serious injury or death.

Note: The following identification is to be used only as a reference only, refer to engineered general arrangement drawing for accurate depiction.

Component Overview: External



1. Hydraulic Opening and Closing
2. Inspection Doors
3. Replaceable Upper Feed Liners
4. Blow Bars
5. Side Liners
6. Bearing
7. Lid Safety Lock

Installation and Commissioning



Read and understand equipment operator's manual before operating or performing maintenance. Failure to do so could result in serious injury or death.

⚠ WARNING

Failure to follow instructions below could result in death or serious injury:

- Lockout/Tagout before performing maintenance and operation.
- Follow installation instructions completely. Failure to do so may cause death or serious injury.
- Have qualified electrician connect power to equipment. Not doing so can result in personal injury and/or death.

Installation

Inspect Equipment

Note: Inspect Equipment prior to installation.

1. According to material processing requirements of machine, configure its feed, discharge chute and protective guard for sheave & motor.
2. Examine whether all internal and external assemblies of machine are fixed.

Note: Check fasteners and tighten in sequence to ensure secure.

Site Preparation

1. Crusher must be placed on solid, level, and blocked foundation.
2. Note: If ground is soft be sure that blocking is large enough to prevent settling

Note: Flatness should not exceed 1 mm.

Note: Level up when installing machine.

Note: Hard rubber pads can be added under main frame for eliminating shocks.

3. Main machine can be mounted on cement concrete foundation or steel structure.

Note: Understand characteristics of foundation soil layer and determine size of pressure zone.

Note: Secondary clay layer or secondary clay layer with fluidity should be noticed.

Note: Foundation levelness should be good enough to enable machine to achieve accurate alignment and levelness.

Note: Lower part of crusher frame is allowed to do secondary grouting treatment.

4. Anchor bolts installed on concrete foundation should be tightened after 72 hours of secondary pouring.

Note: Steel skid structure is required to be smooth and strong enough, with bolt fastening.

Note: Steel structure is required to be solid and structure should be set firmly on a cement concrete foundation.

Note: crusher shall be installed in horizontal lever (flatness <1 mm); otherwise bearings of crusher will run badly.

Note: When making steel structure frame or concrete foundation, configuration of technological process, layout of discharge chute and conveyors layout should be considered.

Motor

1. Motor shall be installed on guide rail so adjusting tightness of belt is convenient.

Note: Rotor shaft, belt sheave, and motor sheave shall be installed horizontally and equipped with a protective cover for transmission belt.

Curtain Liners

1. Clearance between curtain liners and blow bars can be adjusted according to physical characteristics of material and required particle size of crushed product.

On condition that material is hard and discharge material requires a fine gradation, clearance should be smaller.

2. After adjustment turn rotor in several circles to check whether there is any impact or interference.
3. After adjustment is completed, socket nuts of curtain assembly shall be fastened to prevent curtain liners from falling down due to nut-loose caused by vibration from hitting blow bars, resulting in equipment safety accident.
4. Check whether installation location of each component is moved or deformed and all bolts are locked, and seal condition is good.

Electrical

1. Check wire connection and fastening condition of electrical control cabinet; regulate time-delay relay and overload protector. Connect electric circuit to test steering of motor, and select fuse with appropriate specifications.

Hydraulic System

1. Check whether hydraulic system operates normally and reliably and whether there is any leakage.

Chamber

1. Remove foreign matters in chamber of crusher, and turn rotor by hand to check whether there is any interference or collision.

Commissioning

Check before Trial-running

Ensure that all machine parts and components are installed in place and meet operation and safety requirements.

Notice all connection points and protective devices, bolts (thread) etc. are fastened in accordance with specifications.

Close cavities and door covers, etc.

Other staff shall be then required to evacuate away from hazardous operating area

(this step should be taken seriously and many potential safety hazards should be eliminated during this period).

1. Check whether state of bearing lubrication is reasonable.
2. Check whether sheaves are aligned and whether belt is tightened properly, without being too tight or too loose.
3. Check whether electrical control device and other relevant accessories are normal.
4. Check whether rotor *rotates* smoothly by hand and whether there is abnormal sound such as collision sound.
5. Inching-start motor to check whether rotor rotates in right direction.

Note: Machine is strictly prohibited to reverse.

Trial-running without Load

Machine can be started when inspection work is confirmed in readiness before trial-running.

Closely monitor starting time, no-load current and voltage data, and check whether machine is running smoothly, without abnormal vibration and sound, etc.

If there is any abnormality, please stop machine immediately for inspection.

Check whether bearing temperature rise is normal bearing temperature rise should not exceed 1.67°C (35°F); highest temperature of bearing should not exceed 26.67°C (80°F).

Check whether there is any loose with locking fasteners.

No-load running time is 2–4 hours. If everything goes well, it can be put into operation with load.

Trial-running under Load

Start discharge and feed devices in sequence.

Material can be fed only after system operates normally.

Feeding rate should be about 50% of rated passing amount.

Note: Feeding should be evenly aligned, and check and record relevant technical parameters every 30 minutes or 1 hour.

Observe operation of machine for 4–8 hours, and pay attention whether vibration of machine, tightness of triangle belt, wear of vulnerable parts of rotor, motor current, lubrication and other conditions are normal.

After normal operation with 50% of rated load, conduct operation test with 75% of rated load and full load step by step.

At same time, pay close attention to operation of equipment and record relevant technical parameters.

1. Machine should operate steadily without abnormal vibration.
2. Bearing temperature: temperature rise should not exceed 7.2°C (45°F)
3. Machine vibration: vibration at base where bearing locates should be less than 0.2 mm in vertical direction.
4. Voltage: Pay attention to its error, should not operate in under-voltage or significant over-voltage condition.
5. Current: Less than rated current of motor, and pay attention to fluctuation range.
6. Throughput: Pay attention to reasonably weighting and calculating, t/h.
7. At each door and sealing belt, there shall be no dust flying in connection of each sealing place.
8. Wear monitoring of wear-resistant parts.
9. Trial-running with load shall be carried out continuously for 5–8 hours.

Note: Only after running is confirmed normal can machine be put into formal operation.

Operations and Maintenance



Read and understand equipment operator's manual before operating or performing maintenance. Failure to do so could result in serious injury or death.

WARNING

Failure to follow instructions below could result in death or serious injury:

- Lockout/Tagout before performing maintenance and operation.
- Follow installation instructions completely. Failure to do so may cause death or serious injury.

Operations

1. Before starting machine, operators have to check each lubrication point, and make sure that grease is enough, connecting bolts are fixed, and all doors are closed tightly.
2. Check whether clearance between blow bars and curtain liners is appropriate.
3. After starting motor, it is necessary to feed material evenly and gradually after machine operates stably.
4. Material-feeding must be stopped before stopping operation, and motor can only be stopped after all material in crushing chamber is exhausted out.
5. Blow bars should be checked frequently on their wear, and replaced in time to avoid abrasion of rotor body. After blow bars are replaced, static balance test should be carried out.

Maintenance

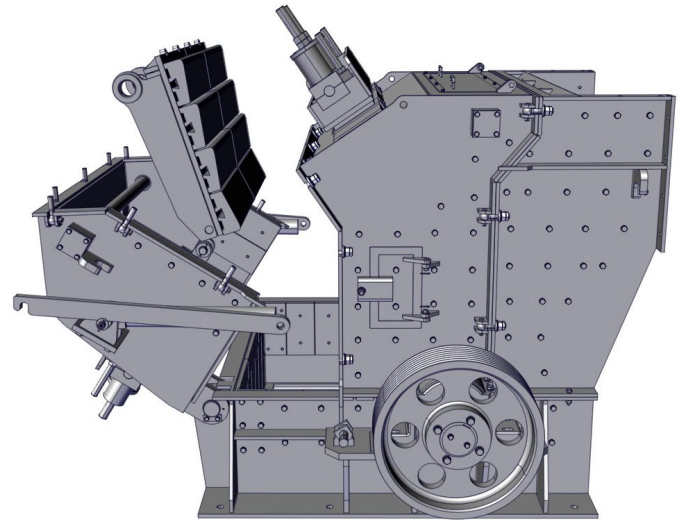
There is an inspection door on both sides of the equipment. Operators can observe interior of machine through inspection ports on frame.

Note: Machine must be stopped to use inspection ports.

When needing to replace blow bars or curtain liners, Hydraulic cylinders on both sides of frame can be raised by hydraulic station, so as to open frame for careful observation and maintenance.

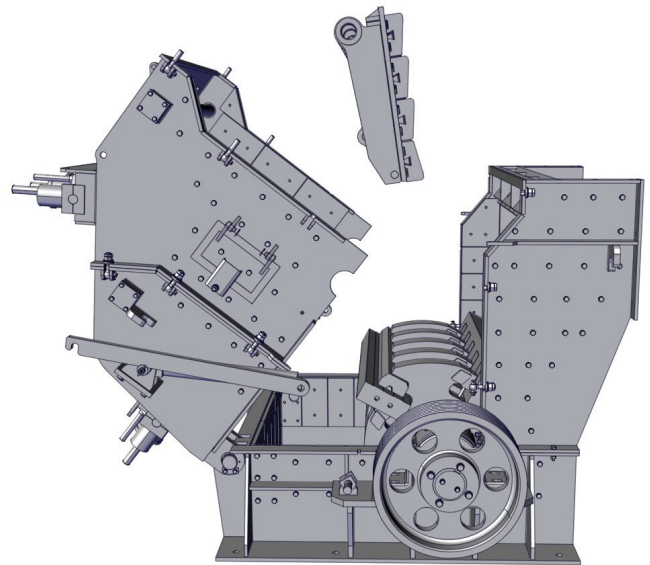
Note: It is forbidden to open covers or inspection doors to check or look into machine during operation.

Figure 5



Opening of Rear Frame Diagram (Figure 5).

Figure 6



Opening of Middle Frame and Rear Frame Diagram (Figure 6).

Open Frame

1. Remove all connecting bolts,
2. Connect hydraulic station to corresponding interface.
3. Jack up grease cylinder.

Note: To avoid safety accident caused by cylinder pressure release, when rear frame is opened reclining to extreme position of limited mechanism at base, have safety rod support at both ends of cylinder at base.

If pressure shown on pressure gauge still rises rapidly after exceeding 10 MPa during operation, please stop pump immediately and check whether there is any accident or interference.

- In process of opening, attention should be paid to avoiding tip-over by self-weight of curtain assembly.
 - In process of closing rear frame, pay attention to operating steadily and keeping alignment.
4. After opening rear frame, replacement and maintenance of secondary curtain liners can be carried out (*Figure 5*).
 5. When rear frame and middle frame are open, blow bars of rotor assembly and liners of primary curtain can be replaced and maintained (*Figure 6*).

Note: Check rotor components after opening middle and rear frame.

Rotor assembly

1. Turn one blow bar upwards and fix rotor with locking pins to prevent free rotation.
2. Remove wedges before replacing or reversing blow bar.
3. After installation of each blow bar, fasten corresponding wedges tightly.
4. Make up worn part of rotor body with wear-resistant electrode as appropriate.
5. Static balance shall be checked after maintenance of rotor or replacement of blow bars.
6. When testing static balance, it is required to remove transmission triangle belts and turn rotor to observe its stop point.
7. If some points always stay downwards, eventually, it can be built up with welding on its opposite side or small iron counterweight.
8. Other components which can be dismantled on rotor shall be weighed and matched before assembly.

Note: That match their weight when replacing blow bars for balance.

Note: Adjustment shall be carried out when frame is closed.

Adjustment of primary curtain

1. Connect hydraulic grease pipes via quick joints to hydraulic cylinders of adjusting mechanism.
2. Remove rod caps on both ends of adjusting base and then adjust clearance setting between curtain liners and blow bars by jacking up hydraulic cylinder.

Note: Rod may break if cap is not removed prior to adjustments.

3. Monitor clearance setting through inspection ports at both sides. After adjustment, re-install and fasten screw rods.
4. Adjustment of secondary curtain: same as that of primary curtain.
5. Replacement of frame liners:
6. Frame liners are retained by bolts.

Note: Pay attention to specification and position of frame liners during replacement and tighten them thoroughly.

Motor operation and maintenance

Replacement and regular cleaning of motor bearing grease.

Unpredictable damages may occur to various parts and components of machine during operation.

Closely observe and timely maintain a suitable cycle of wear parts replacement and maintenance.

Lubrication

- Bearings should use #2 lithium based lubricating grease.
- Grease should be added when fresh grease drops appear around sealing areas.
- Generally grease should be added once every 10 hours of operation.
- Bearing tracks should be fully filled with grease and each bearing should be greased about 1/3 to 1/2 of its inner cavity volume.

Note: Add grease through nipple on top of bearing housings.

- It is better to add grease after 2–4 hours of operation and when bearing temperature is high (bearing temperature may rise after adding fresh grease, but it would fall down gradually and stabilize, please pay attention to monitoring).
- Regularly check grease in bearing housings and clean it as needed.

Note: Dust and debris are strictly prohibited into bearings housings.

Horizontal Shaft Impactor Start-Up Checklist

Starting New Crusher

	Ok	See Notes
Controls & Interlocks		
Interlocks of lube grease pump motor and crusher drive motor have been verified. (ie. Crusher drive motor will not run without lube pump motor running.)		
Suggestion: Test with drive belts off		
Interlock of lid limit switch and drive motor.		
Suggestion: Attempt to start crusher drive motor with lid raised 50.8mm (2").		
Interlock of vibration switch and crusher drive motor.		
Suggestion: Attempt to start crusher drive motor with vibration detector tripped.		
Crusher mounted sensors, hydraulic power unit junction box, and lubrication package junction box wiring complete and tested.		
Superior or customer supplied control system installed and all functions tested.		
Notes:		
Lubrication System		
Document specific brand and type of lubrication grease used: (ISO-VG 150)		
Suggestion: Check lube grease level.		
Pitch of drain line is minimum of 25.4mm (1.00") vertical change for ever 304.8mm (12.00") horizontal change. If vertical drop of more than 1219.2mm (48") , trap is installed.		
Suggestion: Lube unit should be at least 12" below crusher base.		
Lube grease pump motor has been jogged to verify clockwise rotation (as viewed from motor end) and arrow decal on motor matches direction of rotation.		
All crusher grease piping has been adequately flushed. (Filter element may require changing).		
Lube tank and all fitting connections have been checked for leaks.		
Notes:		
Hydraulic System		
Test hydraulic cylinder opening and closing function.		
Notes:		
Mechanical		
Crusher mounting hardware torque checked and to specification. Grouting (if used) is properly cured.		
Crusher drive motor has been jogged to verify counter-clockwise crusher rotation as viewed top.		
Check alignment of crusher and motor sheaves and v-belt tension.		
Record motor position relative to crusher sheave:		
Lock out discharge belt before accessing underside of crusher. Check discharge area for free fall of material to prevent build up of material.		
Notes:		

Starting & Running Crusher

	Ok	See Notes
Lube grease drain line temperature is above 16°C (60°F)		
Run crusher no load until drain line temperature of 27°C (80°F) is reached. Two hours minimum.		
Coast down time is: _____ seconds after "No Load Run"		
Crusher sheave speed is: _____ (Motor Dia. / Crusher Dia. x Motor RPM = Crusher RPM)		
Full load amps of crusher drive motor nameplate is:		
No-load amps of crusher drive motor is:		
Coast down time is: _____ seconds after 80% load run.		
Package lube mounted relief valve stopped by-passing at: _____ °C (_____ °F) drain line temp.		
There is adequate feed distribution and material is hitting center of feed plate.		
Notes:		

After 8 Hour Initial Operation

	Ok	See Notes
Drain line temperature is: _____		
Supply grease temperature is:		
Supply grease pressure is: _____ MPa / psi		
Verify drive belts have been re-tensioned per belt manufacturers instructions.		
Notes:		

Portable Plants

	Ok	See Notes
Plant is level.		
Cribbing is properly installed.		
All lube and hydraulic hoses are adequately supported and protected from rubbing.		
All cabling are adequately supported and protected from rubbing.		
All tires are off ground.		
All guards are properly secured.		
All ladders are secure.		
All safety chains are in place.		
Plant has proper clearance between ground and conveyor.		
Customer has received portable plant manual.		
Notes:		

Comments:

Pre-Operational Checklist



Read and understand equipment operator's manual before operating or performing maintenance. Failure to do so could result in serious injury or death.

WARNING

Failure to follow instructions below could result in death or serious injury:

- Lockout/Tagout before performing maintenance and operation.
- Follow installation instructions completely. Failure to do so may cause death or serious injury.

Pre-Operation Tests

Prior to starting equipment, perform the following tests:

Grease Flow Shutdown Switch

1. Check low grease flow shutdown switch located on grease system for proper function at regular intervals.
2. Attempt to start main motors with grease system off.

Note: If motors start, shut down immediately and contact Superior Industries.

Lid Test

1. Raise lid until limit switch is no longer in contact with shaft assembly.
2. Attempt to start crusher.

Note: If motors begin to start, shut down immediately and contact Superior Industries.

Pre-Operational Checks

Prior to starting equipment, refer to bullet points below:

- Ensure machine is level.
- Ensure foot mounts are bolted down with proper hardware.
- Use metal detector to ensure no foreign metal is introduced into crusher.
- Remove tramp iron and oversized feed from feed conveyors.
- Check for loose/missing bolts.
- Check drive belt for dirt or foreign matter.
- Check belt tension.
- Check grease intervals.
- Check v-belt tension.
- Verify material meets feed/speed recommendations.
- Check rotation direction of rotor (rotation direction arrow is marked on sheave).

Note: Reverse is strictly prohibited.

Common Faults and Trouble Shootings



Read and understand equipment operator's manual before operating or performing maintenance. Failure to do so could result in serious injury or death.

WARNING

Failure to follow instructions below could result in death or serious injury:

- Lockout/Tagout before performing maintenance and operation.
- Follow installation instructions completely. Failure to do so may cause death or serious injury.

Common Faults	Possible Causes	Troubleshooting
Abnormal vibration	• Feed material too large • Uneven abrasion • Rotor unbalance • Foundation problem	• Check feed material size • Replace blow bars • Calibrate rotor frame balance • Check whether anchor bolts are loose
Bearing heating	• Grease shortage • Grease overdose • Bearing broken	• Timely greasing • Check the amount of grease • Replace the bearing
Discharge material particle oversize	• Blow bars abrasion • The clearance between blow bars and curtain liners too large	• Replace the blow bars • Adjust the clearance setting
Abnormal sound in the crushing chamber		• Fasten the bolts • Adjust the clearance • Eliminate foreign matter

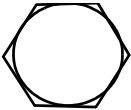
Torque Chart

The following torques should be used when assembly torque for a particular fastener or device is not specified

US Standard

FT/LBS	Grade 2		Grade 5		Grade 8		18-8 S/S		Bronze		Brass	
	Coarse	Fine	Coarse	Fine	Coarse	Fine	Coarse	Fine	Coarse	Fine	Coarse	Fine
1/4"	4	4.7	6.3	7.3	9	10	6.3	7.8	5.7	7.3	5.1	6.4
5/16"	8	9	13	14	18	20	11	11.8	10.3	10.9	8.9	9.7
3/8"	15	17	23	26	33	37	20	22	18	20	16	18
7/16"	24	27	37	41	52	58	31	33	29	31	26	27
1/2"	37	41	57	64	80	90	43	45	40	42	35	37
9/16"	53	59	82	91	115	129	57	63	53	58	47	51
5/8"	73	83	112	128	159	180	93	104	86	96	76	85
3/4"	125	138	200	225	282	315	128	125	104	102	118	115
7/8"	129	144	322	355	452	501	194	193	178	178	159	158
1"	188	210	483	541	682	764	287	289	265	240	235	212

US Bolts

Head Marking	Grade and Material	Nominal Size Range (inches)	Mechanical Properties		
			Proof Load (psi)	Min. Yield Strength (psi)	Min. Tensile Strength (psi)
 No Markings	Grade 2 Low or medium carbon steel	1/4" thru 3/4"	55,000	57,000	74,000
		Over 3/4" thru 1-1/2"	33,000	36,000	60,000
 3 Radial Lines	Grade 5 Medium carbon steel, quenched and tempered	1/4 thru 1"	85,000	92,000	120,000
		Over 1" thru 1-1/2"	74,000	81,000	105,000
 6 Radial Lines	Grade 8 Medium carbon alloy steel, quenched and tempered	1/2" thru 1-1/2"	120,000	130,000	150,000
 A325	Grade A325 Carbon Alloy Steel with or without Boron	1/2" thru 1-1/2"	85,000	92,000	120,000
Stainless markings vary. Most stainless is non-magnetic	18-8 Stainless Steel alloy with 17-19% Chromium and 8-13% Nickel	All sizes thru 1"		20,000 Min. 65,000	65,000 Min. 100,000 - 150,000 Typical

Tightening Torque: K=0.20

Nominal Diameter	Threads per Inch	 ASTM A307 Grade A	 SAE J429 Grade 5	 SAE J429 Grade 8	 FNL Grade 9
Coarse Thread Series					
1/4	20	43 in-lbs	10 1in-lbs	143 in-lbs	168 in-lbs
5/16	18	88	209	295	346
3/8	16	13 ft-lbs	31 ft-lbs	44 ft-lbs	51 ft-lbs
7/16	14	21	49	70	82
1/2	13	32	75	106	125
9/16	12	46	109	154	180
5/8	11	64	150	212	248
3/4	10	113	267	376	441
7/8	9	182	429	606	710
1	8	273	644	909	1065
1-1/4	7	545	1121	1817	2130
1-3/8	6	715	1469	2382	2792
1-1/2	6	949	1950	3162	3706
Fine Thread Series					
1/4	28		116 in-lbs	164 in-lbs	192 in-lbs
5/16	24		231	327	383
3/8	24		35 ft-lbs	49 ft-lbs	58 ft-lbs
7/16	20		55	78	91
1/2	20		85	120	141
9/16	18		121	171	201
5/8	18		170	240	281
3/4	16		297	420	492
7/8	14		474	669	784
1	14		722	1020	1195
1-1/4	12		1241	2012	2358
1-3/8	12		1672	2712	3278
1-1/2	12		2194	3557	4169

Metric

Class 4.6 and 8.8

Nominal Dia. (mm)	Pitch	Class 4.6				Class 8.8			
		Clamp Load (lbs)	Tightening Torque			Clamp Load (lbs)	Tightening Torque		
			Lubricated (ft-lbs)	Zinc Plated (ft-lbs)	Plain & Dry (ft-lbs)		Lubricated (ft-lbs)	Zinc Plated (ft-lbs)	Plain & Dry (ft-lbs)
4	0.7	333	0.7	0.7	0.9	858	1.7	1.9	2.3
5	0.8	538	1.3	1.5	1.8	1387	3.4	3.9	4.5
6	1	763	2.3	2.6	3.0	1968	5.8	6.6	7.7
7	1	1095	2.8	4.3	5.0	2822	9.7	11	13.0
8	1.25	1389	5.5	6.2	7.3	3580	14.1	16	18.8
10	1.5	2200	10.8	12.3	14.4	5671	27.9	31.6	37.2
12	1.75	3197	18.9	21.4	25.2	8240	48.7	55.1	64.9
14	2	4379	30.2	34.2	40.2	11289	77.8	88.1	103.7
16	2	5943	47	53	62	15320	121	137	161
18	2.5	7301	65	73	86	18822	167	189	222
20	2.5	9286	91	104	122	23938	236	267	314
22	2.5	11509	125	141	166	29669	321	364	428
24	3	13372	158	179	211	34471	407	461	543
27	3	17428	232	262	309	44924	597	676	796
30	3.5	21266	314	356	419	54819	809	917	1079
33	3.5	26310	427	484	570	67821	1101	1248	1468
36	4	30982	549	622	732	79866	1415	1603	1886

Tightening Torque: $K=0.20$

Nominal Diameter	Pitch	 Class 4.6	 Class 8.8	 Class 10.9	 Class 12.9
4	0.7	10.5 in-lbs	27 in-lbs	38.7 in-lbs	45.2 in-lbs
5	0.8	21.2	54.6	78.1	91.3
6	1	36.1	92.9	133.0	155.4
7	1	60.3	155.5	223	260
8	1.25	87.5	225.4	323	377
10	1.5	14.4 ft-lbs	37.2 ft-lbs	53.2 ft-lbs	62.2 ft-lbs
12	1.75	25.2	64.9	92.8	108.5
14	2	40.2	103.7	148.4	173.4
16	2	62	161	230	269
18	2.5	86	222	318	372
20	2.5	122	314	449	525
22	2.5	166	428	613	716
24	3	211	543	777	908
27	3	309	796	1139	1331
30	3.5	419	1079	1544	1804
33	3.5	570	1468	2101	2455
36	4	732	1886	2699	3154

Maintenance Log

Serviced By	Work Performed	Date



Warranty

SUPERIOR INDUSTRIES, INC (d/b/a Superior Equipment and as Superior Components)

TERMS AND CONDITIONS

1) OFFER AND ACCEPTANCE. Superior Industries, Inc's ("Seller") acceptance of Buyer's order to purchase products is expressly made conditional on assent to these Terms and Conditions, which along with the Sales Order constitute a binding "Contract" between the parties. This Contract constitutes the complete and final agreement between Seller and Buyer for the products. Any additional or different terms or conditions contained in any document furnished by Buyer, including but not limited to, any purchase order or any acknowledgment, are deemed to be material and are hereby objected to and rejected by Seller. If such agreement shall be deemed an offer or counter-offer by Buyer, Seller expressly rejects such offer or counter-offer and limits acceptance to these Contract terms and expressly objects to any different or additional terms proposed by Buyer. Any actual performance by Buyer or Seller thereafter shall be deemed a renewal of the offer contained in this Contract and acceptance of this Contract without change. In the event of a conflict between the terms of this Contract and the terms of any other document, the terms of this Contract shall control. This offer to purchase Seller's products is valid for thirty (30) days from the date of the Sales Order.

In the event of a significant general material price increase (as measured by any applicable third-party material price index) after the date of issuance of any purchase order, Seller reserves the right to charge a materials surcharge calculated to recoup the actual increase in material cost for the products subject to the purchase order which is attributable to such general material price increase. Documentation of the general material price increase and the actual increase in material cost will be provided to the Buyer upon written request.

2) PAYMENT TERMS. All prices specified in this Contract are FCA Seller's designated location, which constitutes delivery. All risk of damage to or loss of the products from any cause whatsoever shall pass to Buyer upon delivery, even if Seller arranges for shipment of the product. Unless otherwise expressly provided on the reverse hereof, payment shall be made within thirty (30) days from the earlier of the date of delivery or the date of an invoice, without discount. Any discount which may be expressly provided on the reverse hereof applies to the sale price of the products at the shipping point, and does not apply to any charges made for taxes, storage, loading or transportation. All payments shall be made in United States dollars. Interest will be charged at the rate of eighteen percent (18%) per annum, or the maximum interest rate allowable by applicable law, whichever is lower, on all unpaid invoices. Buyer shall pay all taxes and charges of any nature imposed by any federal, state, or local governmental authority by reason of the sale or delivery of the products whether levied or assessed against Seller, Buyer, or the products. Such applicable taxes or charges, if not included in this Contract, shall be invoiced separately. If, in Seller's opinion, reasonable doubt exists as to Buyer's financial condition, Seller may, at any time and without prejudice to any other remedies, suspend or terminate performance of any order, decline to ship, stop any material in transit, or require full or partial payment by Seller in advance.

3) DELIVERY. Any delivery or promise date indicated on the Sales Order is an estimate of the date Seller believes the products will be available for delivery, provided, however, Seller shall not be responsible for any delays in delivery.

4) WARRANTY.

a) New Product(s) Limited Warranty. Subject to the terms, conditions and limitations contained herein, Seller makes the single applicable warranty of paragraphs (i) through (v) below only to the original Buyer of Seller's new product(s). Each specified warranty period begins on the applicable date (the "Warranty Start Date") determined as follows: (1) for Seller-manufactured conveyor components, the date of shipment; (2) for Seller-manufactured capital equipment, the earliest applicable of the following dates, if any: (a) one year from date of shipment if not commissioned, (b) the date of shipment if commissioned and unregistered, or (c) the date of registration if commissioned and registered; or (3) for all other cases, the date of sale. The specified warranty periods are sometimes referred to as "original warranty periods." The terms "shipment" and "sale" as used herein refer to the dates of shipment and sale from by the Seller and not from any distributor of the Seller. The warranties specified below do not apply to any part, portion or component of any product which is manufactured by a third party ("Third- Party Component").

i) Conveying equipment and conveying components (e.g., conveyors and Superior-manufactured conveyor components such as idlers and pulleys) (a) will not fail to operate in accordance with the product(s)' respective specifications due to defects in material or workmanship during the period ending two (2) years from the Warranty Start Date or at 6,000 hours of operation, whichever occurs first, normal wear and tear excluded, and (b) will not incur a failure of the product(s)' respective structural members (e.g., truss) due to defects in material or workmanship during the period ending five (5) years from the Warranty Start Date, normal wear and tear excluded.

ii) Crushing equipment (e.g., cones, jaws and impactors) will not fail to operate in accordance with the product(s)' respective specifications due to defects in material or workmanship during the period ending two (2) years from the Warranty Start Date or 6,000 hours of operation, whichever occurs first, normal wear and tear excluded.

iii) Vibratory equipment (e.g., screens and feeders) will not fail to operate in accordance with the product(s)' respective specifications due to defects in material or workmanship during the period ending two (2) years from the Warranty Start Date or 6,000 hours of operation, whichever occurs first, normal wear and tear excluded.

iv) Washing equipment (e.g., washers and classifiers) will not fail to operate in accordance with the product(s)' respective specifications due to defects in material or workmanship during the period ending two (2) years from the Warranty Start Date or 6,000 hours of operation, whichever occurs first, normal wear and tear excluded.

v) Other Superior Products. Any Superior-manufactured new product to which no single category of subparagraphs (i) through (iv) above clearly applies will not fail to operate in accordance with the product's respective specifications due to defects in material or workmanship during the period ending two (2) years from the warranty start date defined in this paragraph, normal wear and tear excluded.

b) Aftermarket Part(s) Limited Warranty; Exclusion of Third-Party Components. Subject to the terms, conditions and limitations contained herein, Seller warrants only to the original Buyer that Seller's manufactured aftermarket parts (e.g., mainframe, axle, undercarriage) (the "Aftermarket Parts") will not fail to operate in accordance with the Aftermarket Parts' specifications due to defects in material or workmanship during the period length of one (1) year, normal wear and tear excluded. Warranty period starts from date of shipment. The foregoing limited warranty does not apply to any part, portion or component of any Third-Party Component.

c) Coverage Exclusions. The warranties provided in subsections (a) and (b) above shall not apply to any of the following:

i) consumable parts (including but not limited to manganese or other wear parts, screen mesh, v-belts, bearings, oil, liners, etc.);

ii) failure or damage due to improper installation or commissioning;

iii) failure or damage of equipment or equipment support structure (e.g., chassis) due to improper selection or improper support of equipment (e.g., improper cribbing, chassis, etc.);

iv) labor, travel, or freight costs associated with repairs and/or parts installation, unless such costs (1) are associated with a repair or parts installation approved in writing by the Seller, (2) are carried out by or coordinated with a dealer of Seller as may be required by the Seller, and (3) do not exceed a total amount authorized in writing by the Seller;

v) any loss or damages resulting from plant or equipment down time or loss of equipment or plant use, or equipment or plant productivity (all of which include but are not limited to loss or damages resulting from down time or loss of use of the products or components subject to this warranty and any equipment used with such products or components); and

vi) for avoidance of doubt, any of the items excluded in subsection 4(e) below.

d) DISCLAIMER OF ALL OTHER WARRANTIES, EXPRESS, IMPLIED OR STATUTORY. THE LIMITED WARRANTY SET FORTH IN THE FOREGOING PARAGRAPH IS THE SOLE AND EXCLUSIVE WARRANTY WITH RESPECT TO THE PRODUCTS. SELLER MAKES NO OTHER EXPRESS WARRANTY OF ANY KIND OR NATURE AS TO THE PRODUCTS OR THEIR PERFORMANCE EXCEPT FOR THOSE LIMITED WARRANTIES EXPRESSLY SET FORTH IN THE FOREGOING PARAGRAPH AND SPECIFICALLY DISCLAIMS ANY AND ALL REPRESENTATIONS OR WARRANTIES OF ANY KIND OR NATURE CONCERNING THE PRODUCTS, INCLUDING, BUT NOT LIMITED TO, ANY REPRESENTATION OR WARRANTY THAT THE PRODUCTS COMPLY WITH ANY LAW, RULE OR REGULATION. SELLER MAKES NO WARRANTIES WITH RESPECT TO ANY THIRD PARTY COMPONENT AND SELLER SPECIFICALLY SELLS SUCH THIRD- PARTY COMPONENTS "AS IS" WITHOUT ANY WARRANTY. FURTHER, SELLER MAKES NO IMPLIED WARRANTY OF ANY KIND OR NATURE WITH RESPECT TO ITS PRODUCTS OR ANY THIRD-PARTY COMPONENT AND SPECIFICALLY DISCLAIMS ANY AND ALL IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, ANY AND ALL IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, NON-INFRINGEMENT, OR COMPLIANCE WITH ANY FEDERAL, STATE OR LOCAL LAW, RULE OR REGULATION. IN ADDITION, SELLER EXPRESSLY DISCLAIMS TO THE FULLEST EXTENT ALLOWED BY LAW, RULE OR REGULATION ANY WARRANTY PROVIDED UNDER ANY FEDERAL, STATE OR LOCAL LAW, RULE OR REGULATION.

e) Terms and Conditions of Warranty; Voiding of Warranty; Notice Requirements. The limited warranties set forth above shall be null and void if (a) any alterations or modifications are made to a product, (b) a product is not maintained in strict compliance with the maintenance requirements set forth in the maintenance manual for such product or otherwise provided to Buyer of such product, (c) any repairs are made to a product which are not authorized by Seller in writing, (d) any failure of a product to comply with the above limited warranty is not reported to Seller in writing within thirty (30) days of the date such failure first occurs, (e) a product is operated after the failure of any warranty first occurs, (f) a product is used for any purpose other than for the purpose for which it was manufactured, (g) a product is not operated, maintained, and/or repaired in strict compliance with the terms and conditions set forth in any operating manual for the product (including but not limited to exceeding the load bearing capacity of the product), (h) a product is abused or damaged, (i) Buyer fails to deliver the product to Seller for inspection and testing if requested by Seller or Buyer disposes of the product or any part or component on or before the sixtieth (60th) day after sending a written claim under the warranty to Seller, (j) such failure of the limited warranty results from a failure of any Third-Party

Component, (k) Buyer replaces any part or component of the product with a non-authorized part or component other than genuine parts or components sold, distributed or authorized by Seller.

f) Course of Dealing; Course of Performance; Usage of Trade. No course of dealing or course of performance of Seller with respect to the products sold under this Contract or with respect to any of its products to whomever sold and no usage of trade shall be considered in interpreting this Contract or any part thereof and none of the foregoing shall be considered a waiver or modification of any such terms, conditions, disclaimers or limitation of the limited warranties or disclaimers contained in this Contract. No statement, whether written or oral, made by any employee, salesperson, distributor, agent or contractor of Seller which is not set forth in this Contract shall be considered a representation or warranty with respect to any product, its specifications or its performance and all such statements are hereby disclaimed.

g) Exclusive Remedies for Breach of Warranty. The sole and exclusive remedy for any failure of any product to comply with the limited warranty set forth above or any other warranty imposed upon Seller by law, if any, shall, at the election of Seller, in its sole discretion, be either (a) the repair or replacement of the product or component which failed to comply with such warranty or (b) the refund of the purchase price of the product. Except as provided below, any repair or replacement shall carry the same warranty as the original product but only for the remainder of the original warranty period. Buyer's exclusive remedy with respect to any claim arising out of or as a result of Third-Party Component shall be against the third-party manufacturer.

h) Warranty Claims; Notice Requirement; Limited Time to Bring Claims. Any and all claims under the above-limited warranty shall be made to Seller only in writing and not later than thirty (30) days after the date the product first fails to comply with the above-limited warranty but in no event later than the expiration of the original warranty period with respect to which the claim is being made. Any claim under the above-limited warranty made after such period for making a claim shall be null and void. After receiving written notice of the warranty claim, Seller shall determine whether to (a) repair or replace the product or part or (b) refund the purchase price of the product. Seller may require Buyer to return any product or component part thereof which Buyer claims to be defective to Seller at Buyer's cost for inspection as a condition to any claim under the above-limited warranty. No product or part may be returned to Seller without Seller's prior written authorization. If a product which is returned is determined by Seller in its sole discretion not to have failed to comply with the limited warranty, Buyer shall pay costs of removal, repair and/or replacement for such product. If a product which is returned is determined by Seller in its sole discretion to have failed to comply with the limited warranty, Seller shall pay for all repair and/or replacement costs for such product (or refund the purchase price if so elected by Seller) and Seller shall reimburse Buyer for the reasonable costs of shipping the product or component to Seller.

i) Limitation on Liability for Breach of Warranty and Other Claims. If the warranty and the remedy for any failure of any product to comply with any warranty are deemed for any reason to fail their intended purpose, Seller's liability for any failure of any product to comply with any such warranty, together with any and all other liability, if any, arising out of or in connection with such product, including, but not limited to, all claims, whether in contract, tort, or otherwise, arising out of, connected with, or resulting from the manufacture, sale, delivery, resale, repair, replacement, or use of the product, shall not exceed the purchase price for such product. In no event shall Seller be responsible or liable to Buyer or any third party under any circumstances for any indirect, consequential, special, punitive or exemplary, damages or losses, including, but not limited to, damages for loss of profits, goodwill, use of the product or any other equipment or other intangible losses which may be incurred in connection with the product regardless of the type of claim or the nature of the cause of action, even if Seller has been advised of the possibility of such damage or loss. Any and all claims that Buyer has against Seller, whether or not Buyer is aware of such claims, must be brought by Buyer within thirty (30) days after the date that such claim first arose, but in any event within the applicable warranty period set forth above. Any claim not brought by Buyer within the applicable thirty (30) day period shall be deemed null and void.

5) INDEMNIFICATION. Buyer will indemnify and hold harmless Seller, its affiliates and their respective officers, directors, employees, agents and other representatives and defend any action brought against same with respect to any claims, judgments, actions, suits, demands, damages, liabilities, costs or expenses (including, but not limited to, reasonable attorneys' fees and legal expenses) associated with or arising from the ownership, use or operation of the products by Buyer or any third party, including without limitation, product liability, an international, federal or state occupational safety and health statute, or any other governmental regulations or laws.

6) TERMINATION OF PERFORMANCE. Buyer may cancel its order only with the written consent of Seller and upon terms that will indemnify Seller from any loss, damage and expense arising from such cancellation. Seller may terminate this Contract pursuant to Sections 2 and/or 11 hereof, and in such event, Seller shall have no further liability to produce or ship any products hereunder and shall have no liability for damages to Buyer or any third party.

7) TECHNICAL ADVICE. No obligation or liability shall arise out of Seller's rendering of technical advice in connection with Buyer's order of products. Any technical advice furnished, or recommendation made by Seller or any employee or representative of Seller, concerning any use or application of any products or parts furnished under this Contract is believed to be reliable, but Seller makes no warranty, express or implied, of results to be obtained. Buyer assumes all responsibility for loss or damage resulting from the handling or use of any such products or parts in accordance with such technical advice or recommendation. The selection of the products ordered, or design of any custom products, shall be Buyer's sole and ultimate responsibility, and Seller shall have no liability whatsoever for any design defects of custom products, or if the products ordered are unsuitable for Buyer's intended use. Any advice or assistance provided by Seller to Buyer in connection with Buyer's selection or design of the products is at Buyer's risk, and Seller makes no representation or warranty whatsoever in connection with such advice or assistance.

8) BUYER'S REPRESENTATIONS. Buyer represents and warrants to Seller, expressly acknowledging that Seller is relying on these representations and warranties, that: (i) all information supplied by Buyer to Seller in any financial, credit or accounting statement is

and will be true, correct, valid and genuine; (ii) the products are to be used only for business purposes; (iii) Buyer is duly organized, validly existing, and in good standing under the Laws of the state of its formation and is qualified to operate and use the products in the jurisdictions in which the products are used; (iv) Buyer has the full authority to agree to these Terms and Conditions and to perform all of its obligations hereunder.

9) BUYER'S OBLIGATIONS AND COVENANTS. In addition to and not in limitation of any other agreements of Buyer hereunder, Buyer agrees at its sole expense: (a) to use the products in accord with all applicable laws and insurance policies as well as in accord with the best and safest practices recognized in the industry for driving and other operation of the products; (b) to operate and use each of the products within its rated capacity and only for the application for which it was provided, to operate each of the products only in the appropriate environment (including but not limited to weather and/or road conditions) recognized as safe and in accord with best practices in the industry, to follow any instructions, manuals and other materials provided by Seller or the manufacturer of any third-party components utilized in the products, and in the case of damage or defect in the products, to immediately cease operating and using the products until such damage or defect is repaired and immediately report such damage or defect to Seller; (c) to restrict operation of each of the products to safe, careful, competent and trained personnel selected, employed, trained and controlled by Buyer; (d) to properly store the products; (e) that Buyer will not (or permit any party to) misuse or illegally use the products nor permit any of the products; (f) Buyer will not use or permit the use of the products for uses not contemplated or reasonably foreseeable by the Seller; and (g) that Buyer will not alter or permit the alteration of any products without the prior written consent of Seller.

10) ASSIGNMENT. Buyer shall not assign its rights or obligations under this Contract without the prior written consent of Seller, which consent may be withheld for any reason in the sole discretion of Seller. Any attempt at such assignment by Buyer without the prior written consent of Seller shall be deemed null and void. This Contract will be binding upon the parties hereto, and their successors and permitted assigns.

11) SECURITY INTEREST OF SELLER. Title to the products will not pass to Buyer until all required payments have been made to Seller. Until the purchase price and all other applicable costs and expenses are paid in full, Seller reserves a purchase money security interest in the products and the proceeds therefrom, and Seller thereby possesses the rights of a secured party under the Uniform Commercial Code. Upon Seller's request, Buyer shall execute all necessary financing statements and other documents evidencing this security interest with the appropriate state and local authorities. Seller is entitled to and is hereby granted reasonable access to Buyer's locations as necessary to exercise its remedies as a secured party.

12) GOVERNING LAW. This Contract shall be construed, interpreted, and governed by the laws of the State of Minnesota without regard to its conflict of laws principles. The exclusive forum for any disputes arising out of or relating to this Contract shall be any federal or state court sitting in the State of Minnesota. The parties irrevocably consent to such exclusive jurisdiction in such courts and to the proper venue therein.

13) FORCE MAJEURE. Seller does not assume the risk of and shall not be liable for failure to perform any obligation relating to the products caused by civil insurrection, war, fire, strike, labor stoppages or other labor disturbances, acts of God, acts or omissions of Buyer, acts or omissions of the United States Government, floods, epidemics, freight embargoes, shortages of fuel, energy or materials, failure of suppliers or subcontractors to satisfactorily meet scheduled deliveries, or any other cause beyond the reasonable commercial control of Seller.

14) NOTICES. Any notices, consents or other communications required or permitted under this Contract must be in writing and delivered personally, overnight air courier, registered or certified mail or facsimile. Unless otherwise stated in this Contract, notices, consents or other communication will be deemed received (a) on the date delivered, if delivered personally or by facsimile transmission; (b) on the next business day if sent via overnight air courier; or (c) three (3) business days after being sent, if sent by registered or certified mail.

15) SEVERABILITY; WAIVER. The invalidity or unenforceability of any provision of this Contract shall not affect the validity or enforceability of any other provision of this Contract. No waiver of any of the provisions of this Contract shall be deemed, or shall constitute a waiver of any other provision, whether or not similar, nor shall any waiver constitute a continuing waiver. No waiver shall be binding unless executed in writing by the party making the waiver. The Section headings included herein are for the convenience of the parties only and in no way alter, modify, amend, limit or restrict the contractual obligations of the parties.

16) NO THIRD-PARTY BENEFICIARIES; SETOFF. Nothing in this Contract is intended to, or shall, create any third-party beneficiaries, whether intended or incidental, and neither party shall make any representations to the contrary. Seller shall have the right to deduct from any sums it owes to Buyer, any sums or the value of any obligation owed by Buyer to Seller.

17) ENTIRE AGREEMENT. The terms set forth herein constitute the sole terms and conditions of the Contract between Buyer and Seller. Notwithstanding the foregoing or any other term of this Contract, to the extent this Contract conflicts with the terms or conditions of any written distributor agreement between the parties, the written distributor agreement shall control. No other warranty, term, condition or understanding, whether oral or written shall be binding upon Seller, unless hereafter expressed in writing, approved and signed by Seller.

18) SURVIVAL. The provisions of Sections 3, 4, 5, and 7 through 20 shall survive the termination and performance of this Contract.

19) LANGUAGE CLAUSE. The parties acknowledge that the original version of these Terms & Conditions was prepared in English. Any translation of the original English version of these Terms & Conditions into any other language is provided solely as a convenience,

at Seller's sole discretion, or where required by law. If a translation of these Terms & Conditions into any language other than English is prepared or provided by Seller, Buyer, or by any third party for any reason including due to local or national legal requirements, the English version will control to the extent that there is any conflict or discrepancy in meaning between the English version and any translation thereof.

20) SOFTWARE; DATA PLATFORMS. Use of software or data platforms (or use of products incorporating, using or accessing any software or platforms) provided by Superior Industries is subject to the terms of the Superior Industries EULA (available at www.superior-ind.com/EULA) and the Superior Industries Privacy Policy and Terms of Use (available at www.superior-ind.com/Privacy-Policy-Terms-of-Use).

Proposition 65 Warning

WARNING

Breathing diesel engine exhaust exposes you to chemicals known to the state of California to cause cancer and birth defects or other reproductive harm.

- Always start and operate the engine in a well-ventilated area.
- If in an enclosed area, vent the exhaust to the outside.
- Do not modify or tamper with the exhaust system.
- Do not idle the engine except as necessary.

For more information go to: www.P65warnings.ca.gov/diesel

superior-ind.com

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